Lecture Notes on Deep Neural Networks

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1 Fundamentals of Machine Learning

In the first week, we will study fundamentals of Machine Learning. The chapter has following learning objectives:

1. Define AI/ML/DL/DS
2. Explain supervised vs unsupervised learning
3. Understand K-means steps
4. Feature Engineering
5. Compute Manhattan/Euclidean distance
6. Compute accuracy, precision, recall, F1 and interpret threshold tradeoff

1.1 Difference between AI vs. ML vs. DL

AI is a concept to create intelligent machines that can simulate human thinking capability and behavior. It provides the ability to mimic human cognitive thinking such as problem-solving and capability of a computer system to mimic human cognitive behavior and make decisions.

ML is a subset of AI, which provides us statistical tools to explore the data and gives ability to learn. It is a field of study that gives computers the ability to learn through mathematical models without being explicitly programmed.

DL is a subset of ML, which alleviates the need for feature engineering. It utilizes the concept of neural networks to learn and stimulates human brain.

1.2 Fundamental Concepts of Machine Learning

We would like to have a program, which can learn from experience.

The output of Machine learning would be a program.

Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959). Figure 2 illustrates the difference between a traditional program and a machine learning program.

1.3 Applications of Machine Learning

There are several examples of Applications of Machine Learning. These include:

(a) Natural Language Processing
(b) Computer Vision
(c) Computational Biology
(d) Recommender systems by Netflix and Amazon

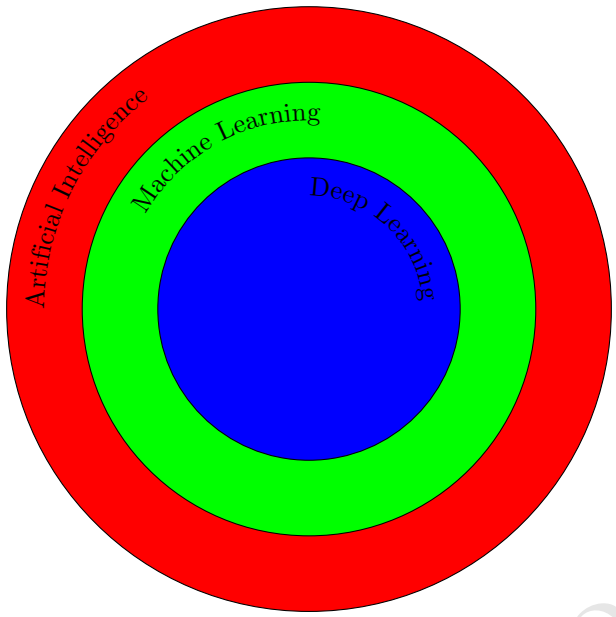

Figure 1: Relationship between AI, Machine Learning, and Deep Learning.

(e) Google ads.
(f) Drug Discovery
(g) Character recognition by Post Office
(h) Assisted Driving

A machine learning model could either have a linear relationship between input and output or it could have been a non-linear relationship.

1.4 Linear vs Non-linear

Linear relationship is modeled through a straight line.

Non-Linear relationship can be represented through any non-linear function For instance, following are examples of non-linear functions:

(a) Quadratic functions: $f(x) = ax^2 + bx + c$
(b) Exponential functions: $f(x) = a \cdot e^x$
(c) Logarithmic functions: $f(x) = \log(x)$
(d) Trigonometric functions: $f(x) = \sin(x),\ f(x) = \cos(x)$

Most of the practical examples in our life are based on non-linear functions. Some of the examples are as follows:

(a) Traffic Flow: Speed of a vehicle on road does not change linearly with the number of other cars on the road
(b) Weight loss: Weight loss or weight gain does not change linearly with exercise
(c) Passing a course: Your grades in a course do not change linearly with the number of hours studied for the course.

1.5 Types of Machine Learning Problems

Machine Learning Problems can be classified as Supervised learning or Unsupervised learning.

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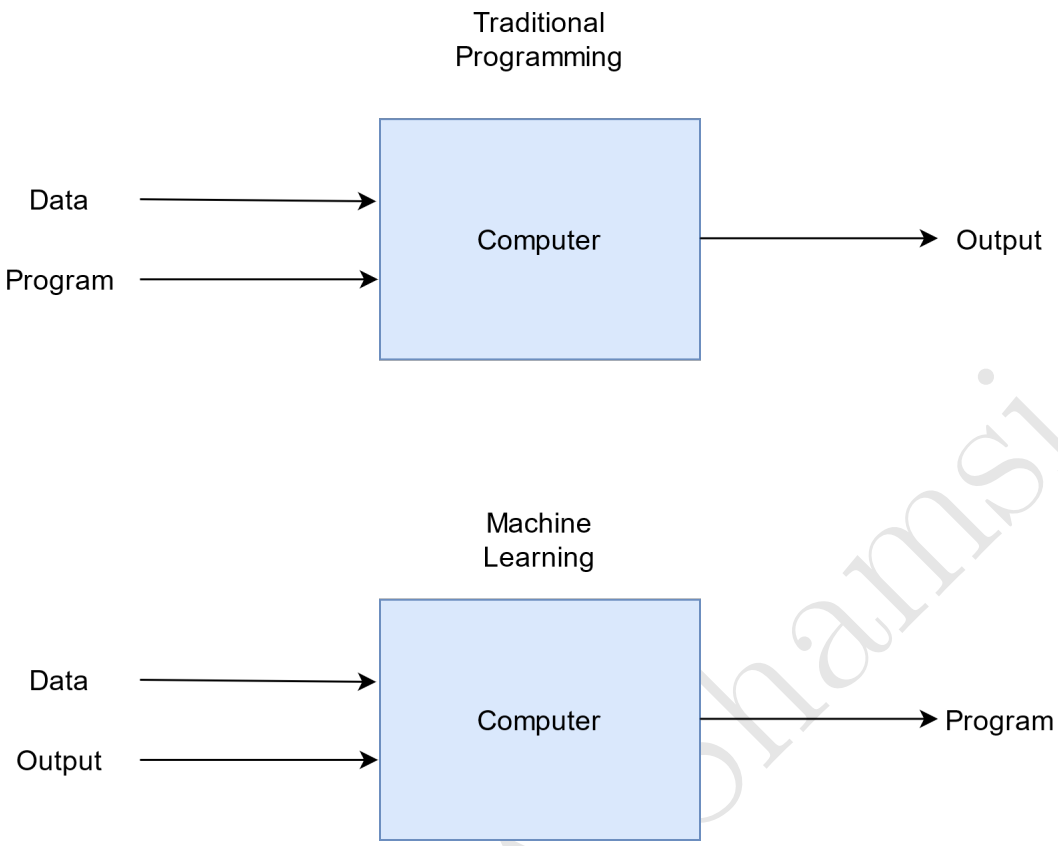

Figure 2: ML vs Prog

1.5.1 Supervised learning

In these problems, data is labeled. In such problems, given a set of features, the task is to find the rule which will predict the label for unseen data.

Types of Supervised Learning

(a) Classification : Predict the class

(b) Regression: Predict the value

Example of Classification : Predict grades based on number of hours studied **Example of Regression**: Predict price of land according to area.

1.5.2 Clustering or Unsupervised learning

In this class of machine learning algorithms, the data is not labeled. Let us group them based on features and see the pattern.

K-Means = A popular clustering algorithm Given a set of unlabeled feature vectors, group them into groups.

(a) Suppose there are K clusters

(b) Pick random k points as center points or centroids

(c) Assign each data points to centroids

(d) cluster remaining samples by minimizing distance between the data points in the same cluster

(e) Recompute each centroid as the mean of the assigned points.

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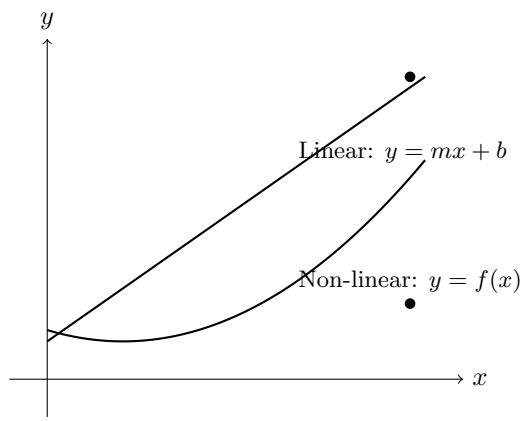

Figure 3: Linear vs Non-linear relationship between input x and output y.

Suppose we have a few data points and we arranged them in to two clusters. Later, we have two new data points, one of them fits well into one of the clusters, whereas the other one is at the edge of the circle. In such a case, we have two options. First, either to re-cluster all the data points or to have more than two clusters.

Let's understand the K-means algorithm through an example: **Step 1**: Input data

Table 1: Data for K-Means

Point	Value
P1	(2,3)
P2	(3,3)
P3	(6,8)
P4	(7,8)
P5	(2,2)
P6	(3,2)
P7	(6,7)
P8	(7,7)
P9	(3,4)
P10	(6,6)

Step 2: Randomly initialize two centroids for k=2. Let's choose:

- **Centroid C1**: (2, 2)
- **Centroid C2**: (6, 8)

Step 3: Compute the distance and assign the points to the nearest centroid. For instance, if Euclidean distance is used then the distance from each point to the centroids can be calculated as follows (first few points):

$$d(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Table 2: Point Distances and Assigned Clusters

Point	Distance to C1 (2,2)	Distance to C2 (6,8)	Assigned Cluster
P1	$\sqrt{(2-2)^2 + (3-2)^2} = 1.0$	$\sqrt{(6-2)^2 + (8-3)^2} = 6.4$	C1
P2	$\sqrt{(3-2)^2 + (3-2)^2} = 1.4$	$\sqrt{(6-3)^2 + (8-3)^2} = 5.8$	C1
P3	$\sqrt{(6-2)^2 + (8-2)^2} = 6.48$	$\sqrt{(6-8)^2 + (8-8)^2} = 2.0$	C2
P4	$\sqrt{(7-2)^2 + (8-2)^2} = 7.8$	$\sqrt{(7-6)^2 + (8-8)^2} = 1.0$	C2

step 4: Compute centroid of each cluster by computing mean of all the points in the cluster

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step 5: Iterate step 3. Stop when the algorithm is converged.

Final cluster assignment

- **Cluster 1**: P1, P2, P5, P6, P9
- **Cluster 2**: P3, P4, P7, P8, P10

1.6 Data

Some questions:

(a) What is the training data and how are we going to evaluate it?

(b) what is the training data and how are we going to evaluate the success of the system?

(c) How do we figure what are the right features and how do we measure distances between them. How to find similarities?

Data can be split into training, testing, and validation:

(a) Training data This refers to the portion of data which is used to train a model. A model train on one set of data (may or) may not perform with similar accuracy on another set of data.

(b) Validation data: The portion of data which is used to assess a model's accuracy or performance during the process of training is called as validation data. The primary purpose is to validate the process of training through unseen data. Training is considered comprehensive (and complete) if the training accuracy is acceptable and is close to the validation accuracy.

(c) Testing Data This refers to the portion of data which is used to assess a model's quality or it's efficiency Never use test data during model design or hyperparameter tuning (to avoid data leakage).

1.7 Features

Features are attributes about data point. Selection of right features are very important in Machine learning.

"All models are wrong but some are useful". George Box

Suppose I would like to predict grades in this course. Features such as prior programming experience, existing CGPA, no. of lectures attended are important. some other features such as color of eyes or residential address may seem irrelevant and may lead to overfitting.

We can input all the features and sorts them to find out which are useful. However, it may lead to overfitting. So goal is to maximize the features which are likely to be useful and minimize those which are not useful.

Let's take an example: We have been given five features and we have to develop a classifier which can identify if a given animal is a reptile or not.

Table 3: Feature Selection Example

Name	Egg-Laying	Scales	Poisonous	Cold-blooded	no. of legs	reptile
Cobra	True	True	True	True	0	Yes
Rattlesnake	True	True	True	True	0	Yes
Boa Constrictor	False	True	False	True	0	Yes
Chicken	True	True	False	False	2	No
Alligator	True	True	False	True	4	Yes
Dart Frog	True	False	True	False	4	No
Salmon	True	True	False	True	0	No
Python	True	True	False	True	0	Yes

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Consider table For the first row, we start by selecting all the five features. These are highlighted in red. This continued for row 2. In row 3, we have a different case. Boa Constrictor does not lay eggs, it is not poisonous and it is a reptile. So we refined the selected features to three. In row 5, we refined the selected features by changing the requirement of no. of legs to either 0 or 4. The problem arises in row 7 and row 8. They both have similar features but the response variable is different. We consider this as an anomaly or outlier. In the final selection, we have reduced the no. of features to two, i.e. scales and Cold-blooded.

Design Choice: No False negatives. That is, no instance, which is a reptile and we call it not a reptile. We may have false positives, i.e., an instance that is not a reptile and we call it as a reptile (example:salmon).

In general, feature selection affects both accuracy and generalization.

1.8 Computing Distance

Distance will be used to group items or how to separate them a part.

Another way to separate reptiles and non-reptiles is to measure the distance between different data points. This is specifically useful if the data is unlabeled.

A simple way is to convert them into feature vectors. Goal is to either classify them through a divider or to cluster them accordingly. Given the data points above, following are the feature vectors of each of the above animals:

(a) Cobra [1,1,1,1,0]

(b) rattlesnake[1,1,1,1,0]

(c) boa constrictor[0,1,0,1,0]

(d) dart frog[1,0,1,0,4]

The question is how to measure distance between these feature vectors? There are two popular methods. That is, Manhattan Distance and Euclidean Distance

1.8.1 Manhattan

Manhattan: Take the sum of the absolute values of the differences of the coordinates.

As per NIST, Manhattan distance is defined as

The distance between two points measured along axes at right angles

For two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, the Manhattan distance is:

Manhattan Distance $= |x_2 - x_1| + |y_2 - y_1|$

Manhattan distance is preferred over Euclidean distance when data is of high dimension.

Example:: Consider two points:

- P(2,3)
- Q(6,8)

The Manhattan distance is calculated as:

$|6 - 2| + |8 - 3| = 9$

1.8.2 Euclidean Distance

For Euclidean distance, we simply compute the square root of the sum of the squares of the differences of the coordinates.

For instance, if x=(a,b) and y=(c,d), the Euclidean distance between x and y is

$$d(x, y) = \sqrt{(a - c)^2 + (b - d)^2} \tag{1}$$

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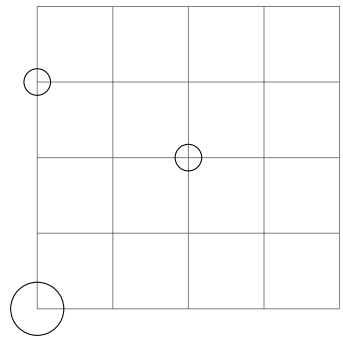

Figure 4: Manhattan Distance.

In general, for n-dimensional data, Euclidean distance can be expressed as follows:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{2}$$

Feature Engineering: choice of features matters too many features may lead to overfitting weights may have an impact

comparing distances using the following feature vector Alligator [1,1,0,1,4] rattlesnake [1,1,1,1,0] boa Constrictor [0,1,0,1,0] dart frog [1,0,1,0,4]

Table 4: Distance Computation

	rattlesnake	boa constrictor	dart frog	alligator
rattlesnake	-	1.414	4.243	4.123
boa Constrictor	1.414	-	4.472	4.123
dart frog	4.243	4.472	-	1.732
alligator	4.123	4.123	1.732	-

In table 4 distance between Alligator and rattlesnake and alligator and boa constrictor is quite high. Although all the three animals are reptiles.

Using binary features in table 5, i.e. an animal has either got legs or no legs, we may get a more realistic value

Table 5: Distance Calculation using binary

	rattlesnake	boa constrictor	dart frog	alligator
rattlesnake	-	1.414	1.732	1.414
boa constrictor	1.414	-	2.236	1.414
dart frog	1.732	2.236	-	1.732
alligator	1.414	1.414	1.732	-

1.8.3 Evaluation Metrics for Classification

In classification problems, the predictions of a model can be summarized using a **confusion matrix**. For binary classification, we define:

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- **True Positive (TP):** Model predicts Positive and the sample is actually Positive.
- **False Positive (FP):** Model predicts Positive but the sample is actually Negative.
- **True Negative (TN):** Model predicts Negative and the sample is actually Negative.
- **False Negative (FN):** Model predicts Negative but the sample is actually Positive.

For binary classification, the confusion matrix summarizes prediction results as follows:

Actual \ Predicted	**Positive (1)**	**Negative (0)**
Positive (1)	True Positive (TP)	False Negative (FN)
Negative (0)	False Positive (FP)	True Negative (TN)

Interpretation:

- **TP:** Correctly predicted Positive
- **TN:** Correctly predicted Negative
- **FP:** Incorrectly predicted Positive (Type-I error)
- **FN:** Incorrectly predicted Negative (Type-II error)

Common Evaluation Metrics:

(a) **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

(b) **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

Precision answers: *"Out of all predicted positives, how many were actually positive?"*

(c) **Recall (Sensitivity):**

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

Recall answers: *"Out of all actual positives, how many did we correctly identify?"*

(d) **F1-Score:**

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

1.8.4 Worked Example: Accuracy, Precision, Recall, and F1

Assume we are building a spam detector where:

- **Positive class** = Spam
- **Negative class** = Not Spam

Suppose we tested our model on 100 emails and obtained the following results:

$$TP = 9, \quad FP = 1, \quad TN = 79, \quad FN = 11$$

Note that the total number of samples is:

$$TP + FP + TN + FN = 9 + 1 + 79 + 11 = 100$$

1) Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{9 + 79}{100} = \frac{88}{100} = 0.88$$

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2) Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{9}{9+1} = \frac{9}{10} = 0.90$$

3) Recall

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{9}{9+11} = \frac{9}{20} = 0.45$$

4) F1-score

$$F1 = 2 \cdot \frac{(0.90)(0.45)}{0.90 + 0.45} = 2 \cdot \frac{0.405}{1.35} = 2 \cdot 0.30 = 0.60$$

1.8.5 Threshold and Precision–Recall Tradeoff

Many classifiers output a probability score $p(y = 1|x)$, where $y = 1$ represents the positive class. We choose a **decision threshold** t:

- Predict Positive if $p(y = 1|x) \geq t$
- Predict Negative if $p(y = 1|x) < t$

Why Precision and Recall trade off:

- If we increase the threshold (make the model strict), fewer samples are predicted as Positive. This usually reduces FP (precision increases), but increases FN (recall decreases).
- If we decrease the threshold (make the model relaxed), more samples are predicted as Positive. This usually reduces FN (recall increases), but increases FP (precision decreases).

1.8.6 Spam Filter Example: Strict vs Relaxed Threshold

Assume we have 100 emails:

- Actual spam emails = 20
- Actual not-spam emails = 80

Case A: Strict threshold ($t = 0.9$) The model predicts only 10 emails as spam:

$$TP = 9,\ FP = 1,\ FN = 11,\ TN = 79$$

$$\text{Precision} = \frac{9}{9+1} = 0.90, \qquad \text{Recall} = \frac{9}{9+11} = 0.45$$

This gives **high precision** but **low recall**. In practice, this means: "When the filter says spam, it is usually correct, but many spam emails still appear in the inbox."

Case B: Relaxed threshold ($t = 0.3$) Now the model predicts 30 emails as spam:

$$TP = 18,\ FP = 12,\ FN = 2,\ TN = 68$$

$$\text{Precision} = \frac{18}{18+12} = 0.60, \qquad \text{Recall} = \frac{18}{18+2} = 0.90$$

This gives **high recall** but **lower precision**. In practice, this means: "Most spam is caught, but some normal emails may be incorrectly moved to the spam folder."

Conclusion:

- **Strict threshold** $\Rightarrow$ high precision, low recall
- **Relaxed threshold** $\Rightarrow$ high recall, lower precision

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1.9 Class Imbalance

A class imbalance problem occurs if we have imbalance of occurrence of classes. For instance, in case of a cancer detection dataset we have fewer instances of cancer. Class imbalance occurs when the distribution between the two classes is highly skewed. Normally, we have more negative instances as compared to the positive instances.

1.9.1 Choosing the right set of parameters

Accuracy can be misleading on imbalanced data. For instance, consider if we have a highly skewed data such as only one positive per hundred negatives. If we develop a model, which always predicts a negative value, such a model will give us 99% accuracy. However, the model will not be acceptable. Accuracy is not reliable when classes are imbalanced.

The failure of accuracy as a metric on imbalanced data is well-known. Consider the case of a dataset with the ratio of 1 positive per 100 negatives. On this task, a model that predicts all cases to be negative yields an accuracy of 99%. This model, nevertheless, is a dummy classifier that always predicts the majority class.

1.10 Summary

We have studied basics of Machine Learning. These include initial concepts, feature Engineering, K-Means, and Distance Calculation

Exercises

(a) **AI vs ML vs DL:** Give **one real-life example** of each:
   i. Artificial Intelligence (AI)
   ii. Machine Learning (ML)
   iii. Deep Learning (DL)

(b) **Linear vs Non-linear:** For each function below, state whether it is **linear** or **non-linear**.
   i. $y = 3x + 2$
   ii. $y = x^2 + 1$
   iii. $y = \log(x)$
   iv. $y = 5$

(c) **Supervised vs Unsupervised:** For each problem below, identify whether it is **supervised learning** or **unsupervised learning**.
   i. Predicting house price using area and location
   ii. Grouping students into clusters based on study habits
   iii. Detecting spam emails
   iv. Finding customer segments from purchase history

(d) **Manhattan Distance:** Compute the Manhattan distance between the following points:

$$P(1, 2), \quad Q(6, 5)$$

(e) **Euclidean Distance:** Compute the Euclidean distance between the following points:

$$P(1, 2), \quad Q(6, 5)$$

(f) **Confusion Matrix → Evaluation Metrics:** A binary classifier is tested on 50 samples and produces the following results:

$$TP = 12, \quad FP = 3, \quad TN = 30, \quad FN = 5$$

Compute:

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i. Accuracy
ii. Precision
iii. Recall
iv. F1-score

(g) **Precision vs Recall (Conceptual):** In a **cancer detection system**, which metric is more important and why?

i. Precision
ii. Recall

(h) **Threshold Tradeoff:** A classifier outputs a probability score $p(y = 1|x)$ and predicts the positive class when:

$$p(y = 1|x) \geq t$$

If the threshold is increased from $t = 0.3$ to $t = 0.9$, explain what happens to:

i. False Positives (FP)
ii. False Negatives (FN)
iii. Precision
iv. Recall

(i) **Class Imbalance:** A dataset contains 1000 samples:

- 990 samples belong to the negative class
- 10 samples belong to the positive class

A model predicts **all samples as negative**.

i. What is the accuracy of the model?
ii. Why is this model not useful in practice?
iii. Which metric(s) would better reflect this failure?

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